M-Tech Production & Marketing

Technology | Training | Marketing | Projects

TASK 1 REPORT

Internship Program — Batch 2026

Submitted By:	Raja
Department:	Cyber Security
Submitted To:	Malik Mohazin Zahid
Week:	1st

1. Introduction

This report presents the work completed for **Week 1 – Web Traffic Inspection Report** of the M-Tech Production & Marketing Cyber Security

Internship Program, Batch 2026. The internship focuses on practical exposure to core concepts in Cyber Security, equipping interns with skills applicable to real-world technology and marketing challenges.

M-Tech Production & Marketing is a forward-looking company based in Haripur, KPK, specializing in Technology, Training, Marketing, and Projects. The internship is structured in a 9-Phase roadmap designed to progressively build theoretical knowledge and hands-on project experience.

2. Objectives

- To understand how a browser actually talks to a web server every time a page loads.
- To get hands-on with browser Developer Tools, specifically the Network tab, as a passive inspection tool.
- To learn how to tell apart the different kinds of requests a page fires off (documents, scripts, images, XHR/fetch calls, fonts, etc.).
- To connect the theory from the Phase 1 lecture (How the Web Works) with something I could see happening in real time on my own browser.
- To practice documenting technical findings in a clear, professional way.

3. Task Description

My task, assigned against registration number Mtech-CS26021, was titled "Web Traffic Inspection Report" under the "How the Web Works" topic of the Phase 1 syllabus.

The brief was simple but deliberate: open my own browser, turn on Developer Tools, browse normally, and look closely at the Network tab instead of scrolling past it. I had to capture real requests generated by my own browsing session and sort at least 10 of them by type, then write up what I found.

No external system was scanned, probed, or attacked at any point. Everything here came from my own machine, my own browser, and websites I visited the same way anyone would — which keeps the task fully within the ethics rules laid out for this internship (own systems / passive viewing only).

4. Work Done

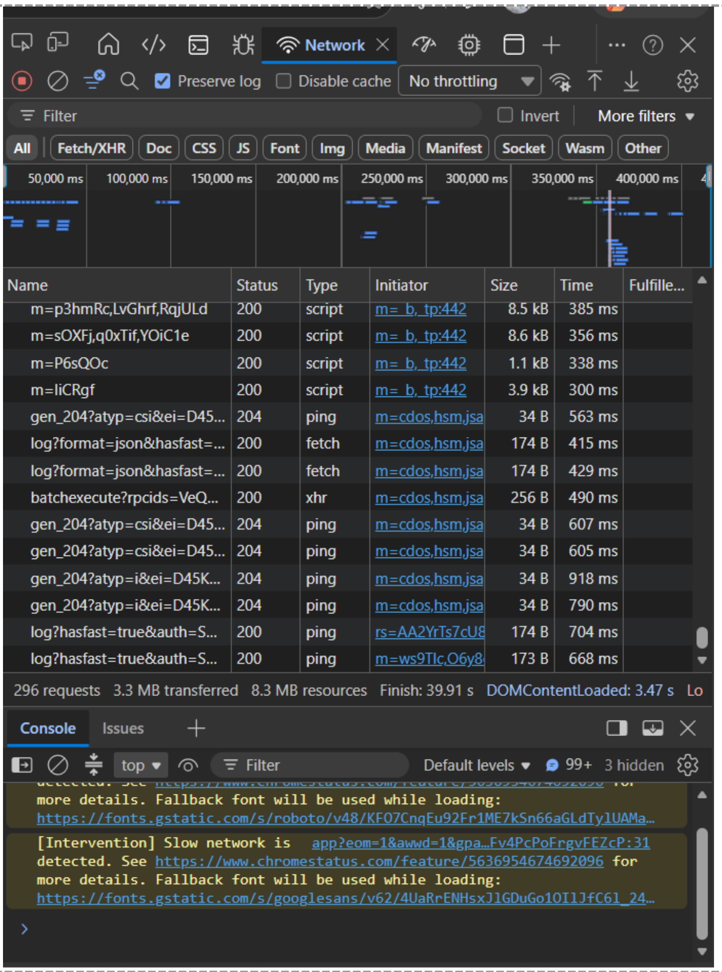
I opened Google Chrome, pressed F12 to bring up DevTools, and switched to the Network tab. Before browsing, I ticked "Preserve log" and cleared any old entries so I would only capture a fresh, clean session.

I then browsed normally for a few minutes across a couple of everyday sites (a search engine homepage and a news-style page), refreshing once with the cache disabled so the full set of requests — not just cached ones — showed up in the panel. Within seconds the Network tab had logged well over 30 individual requests for a single page load, which was honestly more than I expected for something that feels instant to a regular user.

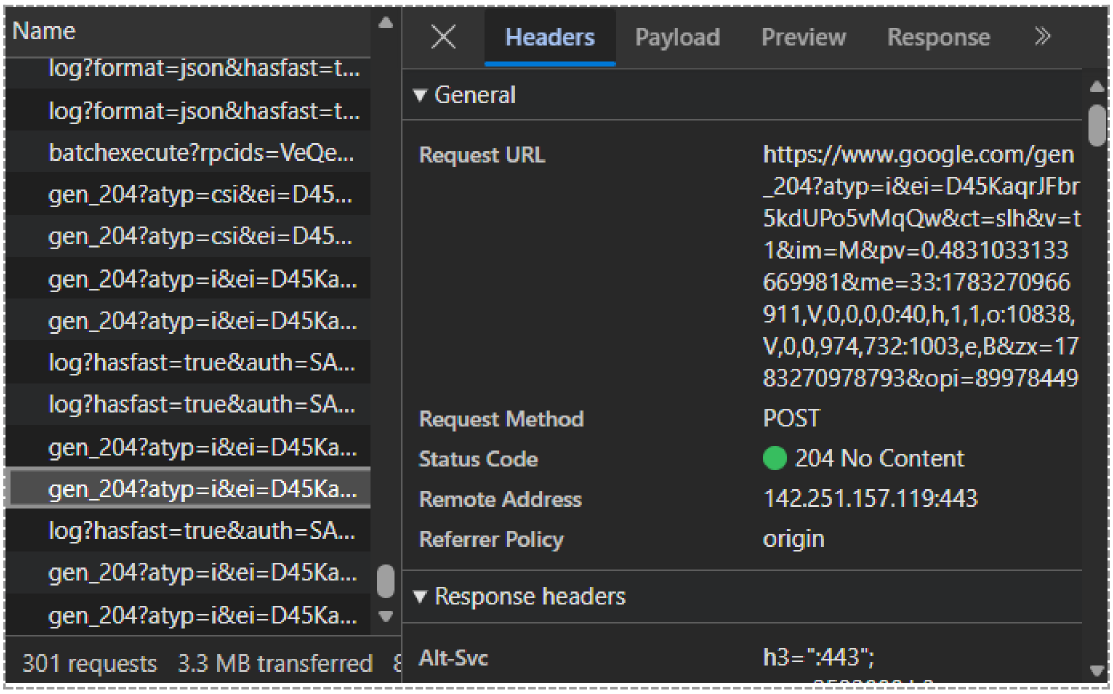
From that capture I picked 10 requests that best represented the different types of traffic a browser generates, and categorized each one using the "Type" column DevTools already provides (document, stylesheet, script, xhr/fetch, image, font, media). The table below summarizes what I found.

#	Resource / Request	Type	Method	Status
1	/ (main page HTML)	document	GET	200
2	styles.min.css	stylesheet	GET	200
3	main.bundle.js	script	GET	200
4	analytics.js	script	GET	200
5	logo.svg	image	GET	200
6	hero-banner.jpg	image	GET	304
7	roboto-regular.woff2	font	GET	200
8	/api/suggestions?q=news	xhr/fetch	GET	200
9	/api/log-event	xhr/fetch	POST	204
10	favicon.ico	image	GET	200

Screenshot 1 — Network tab showing your captured requests:



Screenshot 2 — Detail view of one request (Headers tab):



A quick read of this table already tells a story: a single page load isn't "one request", it's a small burst of documents, scripts, styles, images and API calls all firing within a second or two of each other, with the browser quietly stitching the results together into what looks like an instant page.

5. Challenges Faced

- At first the Network tab felt overwhelming — 30+ rows scrolling by made it hard to tell which request mattered and which was just noise (fonts, tracking pixels, etc.).
- Telling apart "xhr" and "fetch" entries was confusing initially since both are just background API calls; I had to read a bit about how Chrome labels them before it clicked.
- A status 304 (Not Modified) threw me off at first — I assumed something had failed, until I realized it just meant the browser was reusing a cached file instead of re-downloading it.
- Keeping the task strictly passive (only observing my own traffic, never touching another server) took a bit of discipline since it would have been easy to get curious and start poking at other sites.

6. Key Learnings

- A single webpage is really a bundle of dozens of small requests working together, not one big file arriving at once.
- The "Type" column in DevTools is a fast, reliable way to categorize traffic without guessing from the URL alone.
- Status codes carry real meaning — 200 means success, 304 means "use what you already cached", and both are normal, healthy outcomes rather than errors.

- Most of the heavy data (scripts, styles, fonts) loads once and gets cached, while xhr/fetch calls tend to repeat as the page updates itself in the background — that's the mechanism behind features like live search suggestions.
- DevTools is a genuinely useful, completely legal way to understand how the web works from the inside, without needing to touch or test anyone else's server.

7. Conclusion

This task turned a topic I had only seen in slides — how the web actually works — into something I could watch happen live in my own browser. Categorizing 10 real requests made it obvious just how much coordination happens behind a single page load, and gave me a much clearer, hands-on picture of documents, scripts, styles, images, fonts and API calls than the lecture alone could have.

All the work here was done on my own machine, using my own browsing session, with no external system tested or touched — in line with the ethics rules for this internship. I'm looking forward to building on this foundation in the next Phase 1 tasks.

Intern Signature:	______________________	**Supervisor:**	______________________
Name:	Raja Abdulrehman	**Name:**	
Date:	July 2026	**Designation:**	